

Part I (Sit-in)

Simulate a system of N coupled phase oscillators given by:

$$\dot{\theta}_i = \omega_i + K \sum_{j=1}^N A_{ij} \sin(\theta_j - \theta_i) \quad i = 1, \dots, N$$

where local frequencies ω_i are drawn from a Gaussian probability distribution centered about zero with unit variance. A is the Adjacency Matrix.

Investigate the synchronization phenomena arising in the system above, on **Watts-Strogatz Small World Networks**. The base lattice (a ring) from which the WS Small-World network is constructed, has degree 6, i.e. each node is connected to 3 neighbours on each side.

Note: In this part you have to explore **only** synchronization properties.

Part II (Take-home)

Explore the dynamics of coupled maps, where the dynamics of the individual local maps has the following form:

$$x_{n+1} = x_n^2 \exp(r - x_n)$$

where $r \in [1 : 3]$ is a parameter and $x > 0$.

You are free to explore any aspect of the above system that interests you.

The only constraint is that you must use the dynamics given above for the local system at the nodes, though you are free to couple these local units in any manner you choose.

General instructions

1. The parts have equal weightage.
2. You have to upload your work on Moodle (as before), under Module 4 - Part I and Part II - separately. The first part is due by 8:00 pm. The second part is due by 11:00 pm.
3. Also, as in the last 3 modules, you can upload a link to your notebooks (supplemented with a pdf of the main results, if you so wish), or a pdf file of the code and the results.
4. Though open resources in general, you are **not** allowed to consult any form of "natural intelligence" (such as fellow students), and such consultation will be considered a breach of ethics.